Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**AIM:Design FIR filter with windowing method.**	
Experiment No: 06	**Date:24/10/2026**	**Enrollment No:92400133041**

AIM: Design FIR filter with windowing method.

Theory:

The windowing method is a commonly used technique for designing FIR filters. It involves designing an ideal frequency response and then applying a window function to obtain a practical FIR filter.

The steps involved in designing an FIR filter using the windowing method are as follows:

Specify the desired filter specifications, such as the cutoff frequency, filter length, and window type.

Design an ideal frequency response that meets the desired specifications.

Apply a window function to the ideal frequency response to obtain the filter coefficients.

Normalize the filter coefficients to ensure stability.

There are various window functions available, such as the rectangular, Bartlett, Hann, Hamming, and Blackman windows, among others. The choice of window function depends on the desired trade-off between the main lobe width and sidelobe suppression.

Programs:-

1. Design FIR Low pass filter having cutoff frequency 3 KHz, N=31; Sampling frequency=10 KHz

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz

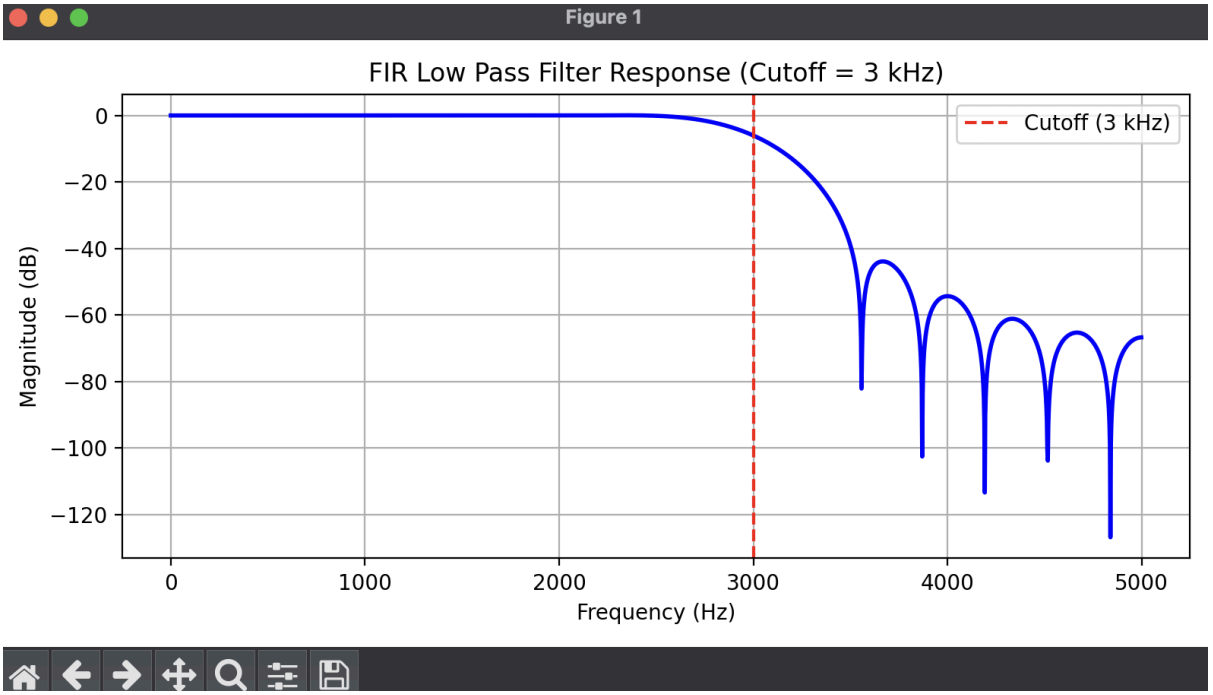
Fs = 10000  # Sampling frequency (10 kHz)
Fc = 3000   # Cutoff frequency (3 kHz)
N = 31      # Number of taps

# 2. Design Low-Pass FIR Filter
b = firwin(numtaps=N, cutoff=Fc, window='hann', pass_zero='lowpass', fs=Fs)
```

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```
# 3. Calculate Frequency Response
f, H = freqz(b, worN=2048, fs=Fs)
magnitude_db = 20 * np.log10(np.maximum(np.abs(H), 1e-10))

# 4. Plot Frequency Response
plt.figure(figsize=(8, 4))
plt.plot(f, magnitude_db, 'b', linewidth=2)
plt.title('FIR Low Pass Filter Response (Cutoff = 3 kHz)')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude (dB)')
plt.grid(True)
plt.axvline(Fc, color='r', linestyle='--', label='Cutoff (3 kHz)')
plt.legend()
plt.tight_layout()
plt.show()
```



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2. Design FIR High pass filter having cutoff frequency 3 KHz, N=31; Sampling frequency=10 KHz

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz

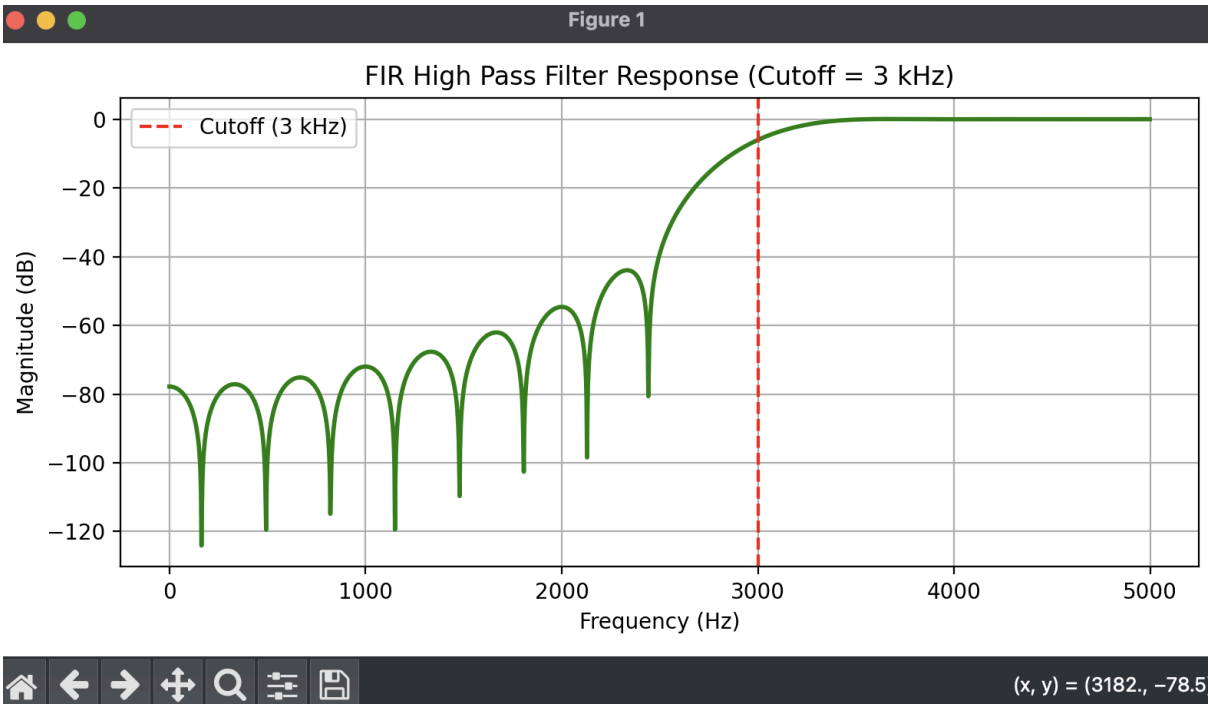
Fs = 10000  # Sampling frequency (10 kHz)
Fc = 3000   # Cutoff frequency (3 kHz)
N = 31      # Number of taps

# 2. Design High-Pass FIR Filter
b = firwin(numtaps=N, cutoff=Fc, window='hann', pass_zero='highpass', fs=Fs)

# 3. Calculate Frequency Response
f, H = freqz(b, worN=2048, fs=Fs)
magnitude_db = 20 * np.log10(np.maximum(np.abs(H), 1e-10))

# 4. Plot Frequency Response
plt.figure(figsize=(8, 4))
plt.plot(f, magnitude_db, 'g', linewidth=2)
plt.title('FIR High Pass Filter Response (Cutoff = 3 kHz)')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude (dB)')
plt.grid(True)
plt.axvline(Fc, color='r', linestyle='--', label='Cutoff (3 kHz)')
plt.legend()
plt.tight_layout()
plt.show()
```

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3. Design FIR Band pass filter having cutoff frequency(3 KHz, 6 KHz) N=31; Sampling frequency=10 KHz

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz

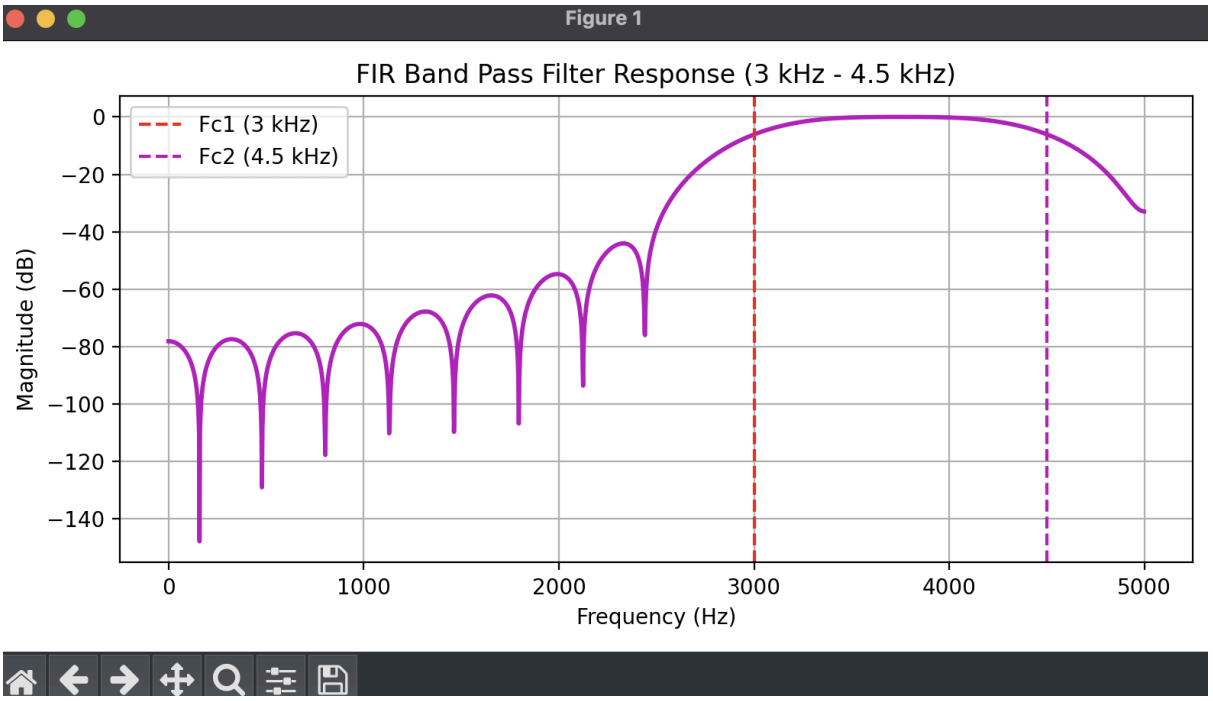
Fs = 10000          # Sampling frequency (10 kHz)
Fc = [3000, 4500]   # Bandpass cutoff range (3 kHz to 4.5 kHz)
N = 31              # Number of taps

# 2. Design Band-Pass FIR Filter
b = firwin(numtaps=N, cutoff=Fc, window='hann', pass_zero='bandpass', fs=Fs)

# 3. Calculate Frequency Response
f, H = freqz(b, worN=2048, fs=Fs)
magnitude_db = 20 * np.log10(np.maximum(np.abs(H), 1e-10))
```

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```
# 4. Plot Frequency Response
plt.figure(figsize=(8, 4))
plt.plot(f, magnitude_db, 'm', linewidth=2)
plt.title('FIR Band Pass Filter Response (3 kHz - 4.5 kHz)')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude (dB)')
plt.grid(True)
plt.axvline(Fc[0], color='r', linestyle='--', label='Fc1 (3 kHz)')
plt.axvline(Fc[1], color='m', linestyle='--', label='Fc2 (4.5 kHz)')
plt.legend()
plt.tight_layout()
plt.show()
```



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4. Design FIR Band stop filter having cutoff frequency(3 KHz, 6 KHz) N=31; Sampling frequency=10 KHz

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.signal import firwin, freqz

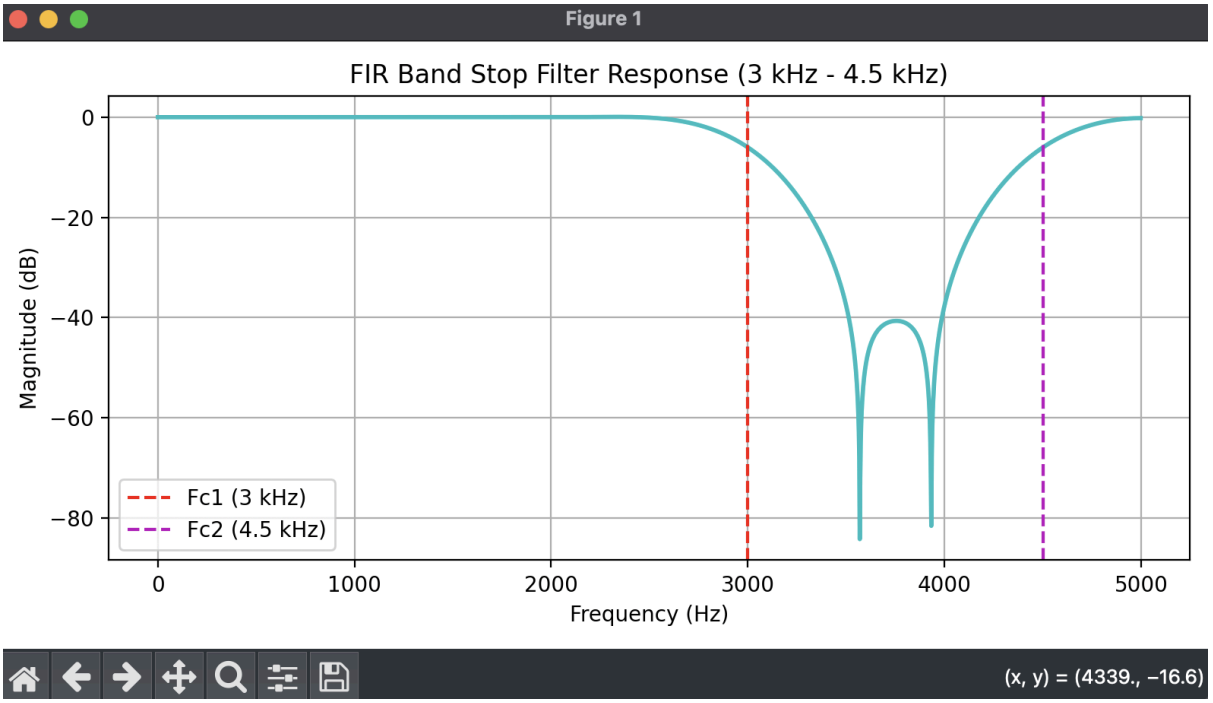
Fs = 10000          # Sampling frequency (10 kHz)
Fc = [3000, 4500]   # Bandstop cutoff range (3 kHz to 4.5 kHz)
N = 31              # Number of taps

# 2. Design Band-Stop FIR Filter
b = firwin(numtaps=N, cutoff=Fc, window='hann', pass_zero='bandstop', fs=Fs)

# 3. Calculate Frequency Response
f, H = freqz(b, worN=2048, fs=Fs)
magnitude_db = 20 * np.log10(np.maximum(np.abs(H), 1e-10))

# 4. Plot Frequency Response
plt.figure(figsize=(8, 4))
plt.plot(f, magnitude_db, 'c', linewidth=2)
plt.title('FIR Band Stop Filter Response (3 kHz - 4.5 kHz)')
plt.xlabel('Frequency (Hz)')
plt.ylabel('Magnitude (dB)')
plt.grid(True)
plt.axvline(Fc[0], color='r', linestyle='--', label='Fc1 (3 kHz)')
plt.axvline(Fc[1], color='m', linestyle='--', label='Fc2 (4.5 kHz)')
plt.legend()
plt.tight_layout()
plt.show()
```

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Conclusion:

This experiment successfully demonstrated the design and realization of Finite Impulse Response (FIR) digital filters—specifically Low-Pass, High-Pass, Band-Pass, and Band-Stop configurations—using the windowing method in Python. By using the `scipy.signal.firwin` function, ideal impulse responses were generated and windowed to control stopband attenuation and sidelobe levels. The frequency response plots generated via Matplotlib confirmed that each filter effectively attenuated frequencies outside its designated passband, illustrating the linear phase stability inherent to FIR filter designs. Overall, the experiment highlighted the practical application of windowing functions for shaping digital frequency responses while maintaining system stability and eliminating phase distortion.

GITHUB LINK:-

https://github.com/SlockAhuja/DSIP.git